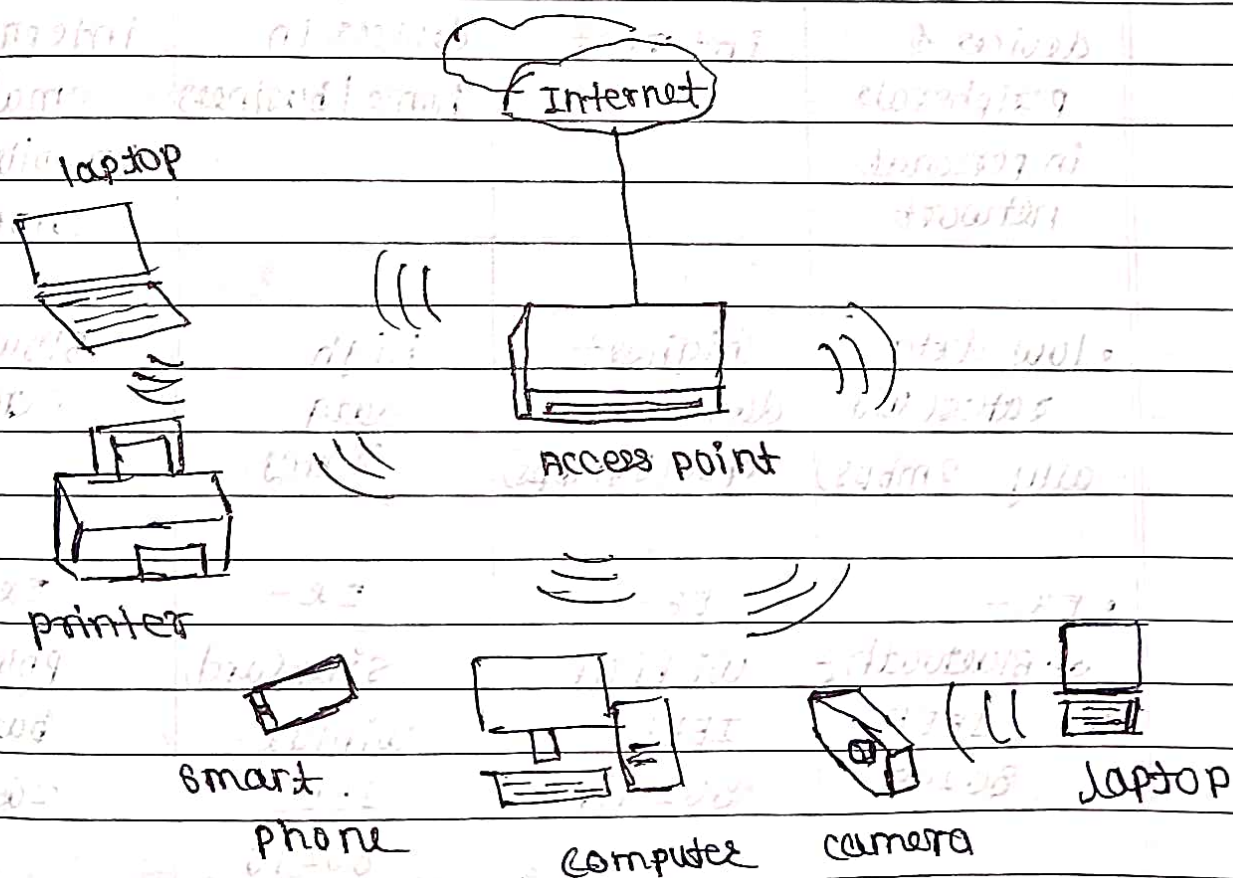


UNIT 03] WIRELESS COMMUNICATION

3-1] Overview of wireless network

- A wireless network connects computers without using network cables.
- They generally use radio waves for communication between the network nodes.
- They allow devices to be connected to the network while roaming around within the network coverage.



* Types

Wireless networks

WPAN

- Works in short range of 50 Feet
- Used to connect devices & peripherals in personal network
- low data rates (usually 2mbps)
- Ex - Bluetooth, IEEE 802.15

WLAN

- Works within building
- Used to connect internet
- highest data rates upto (54mbps)
- Ex - Wi-Fi or IEEE 802.11,

WMAN

- Used to connect two different points in city / town
- Used to connect devices in home / business
- High data rates
- Ex - sim card, wimax- IEEE 802.16

WWAN

- cover large areas ex. country, continent etc.
- used to access internet, emails (mobile phone communication)
- slow network
- Ex - phones based on 2G, 3G,

2.2] IEEE standards :-

802.1, 802.2, 802.3, 802.4, 802.5

- IEEE 802 is a collection of networking standards that cover the physical and data link layer specifications for technologies such as Ethernet & wireless. These standards are apply to LAN and WAN.

• IEEE - Institute of Electrical & Electronics Engineers

- The Family of IEEE 802 is developed & maintained by the IEEE 802 LAN/MAN standard committee also called the LMSC.

Standard	Purpose
i) 802.1	Internetworking
ii) 802.2	Logical link control
iii) 802.3	Ethernet LAN
802.4	Token-Bus LAN
802.5	Token-Ring LAN
802.6	Metropolitan Area Network
802.7	Broadband Technical Advisory Group
802.8	Fiber-Optic Technical Advisory
802.9	Integrated voice OR Data Network
802.10	Network security
802.11	Wireless Networks
802.12	Demand Priority Access LAN

802.13	Not used
802.14	Cable Broadband LANs & MANs
802.15	Wireless Personal Area Networks
802.16	Broadband Wireless Access
802.17	Resilient Packet Rings
802.18	Radio Regulatory TAG
802.19	Wireless Coexistence Working Group
802.20	Mobile Broadband Wireless Access
802.21	Media Independent Handover Services Working Group
802.22	Wireless Regional Area Networks

• IEEE 802.1 :-

• IEEE 802.1 handles the architecture, security, management and inter networking of LAN, MAN & WAN.

• IEEE 802.1 is link with the 802 link security

• It is link with the overall network management of 802

• The IEEE 802 LAN/MAN standards Committee makes current standards freely available after a six month delay, through their Get IEEE 802.1

program.

- Recent 802.1 standards are available through the IEEE for a free.

IEEE 802.2 :-

- It defines logical link control (LLC) as the upper portion of the data link layer of the OSI model.

It was adopted by ISO in 1998. but it remains an integral part of the family of IEEE 802 standards for LAN & MAN.

- LLC is a software component that provides a uniform interface to the user of the data link service, usually the network layer.

- IEEE 802.2 sublayer adds some control info to the msg. created by the upper layer & passed to the LLC for transmission to another node on the same data link.

802.3

- The IEEE 802.3 standard determines the CSMA/CD access control protocol

Uses bus topology

Size of the frame format in IEEE 802.3 is 1572 bytes

No priority given in this

Size of the data field is 0-1500 bytes

Minimum frame required is 64 bytes

Modems are not required

802.4

describes a token bus LAN standards

Tree topology

8202 bytes

Supports priorities to stations

0 to 8182 bytes

It can handle short minimum frames

Modems are required

802.5

describes the token ring standards

Ring topology

Available in variable size

Priorities are possible

No limit on the size of data

It supports both short & large frames

Modems are required

Protocol is
very simple

Protocol is
extremely
complex

Protocol is
moderately
complex

not applicable
on real time
applications

It is
applicable to
real time
traffic

Real time
applications
& interactive
applications.